

Luwei Yang

Physical Oceanographer, Postdoctoral Research Fellow
Research School of Earth Sciences, Australian National University

Academic Profile
luwei.yang@anu.edu.au

Education

- 2015–2019 Ph.D. in Quantitative Marine Science, University of Tasmania, Hobart, Australia.
2011–2015 B.Sc. in Marine Science, Ocean University of China, Qingdao, China.

Professional Experience

- 2023–present Postdoctoral Research Fellow, Research School of Earth Sciences, Australian National University, Canberra, ACT, Australia.
2020–2023 Postdoctoral researcher, Department of Atmospheric & Oceanic Sciences, UCLA, Los Angeles, California, United States.

Publications

- [8] **Yang, L.**, Shakespeare, C.J., B.K. Arbic: Propagating internal-tide-induced wave stresses resolve discrepancies in ocean surface tide energetics. *Geophysical Research Letters*, under review. [preprint]
- [7] **Yang, L.**, C.J. Shakespeare, A.K. Morrison, A.M. Hogg, A.H. Gibson, and B.K. Arbic, 2026: Sensitivity of M_2 barotropic tide solutions to resolution and a physically based wave drag parameterization. *Journal of Advances in Modeling Earth Systems*, 18, e2025MS005528, doi: 10.1029/2025MS005528
- [6] Barnes, A.J., N.C. Constantinou, A.H. Gibson, A.E. Kiss, C. Chapman, J. Reilly, D. Bhagtani, **L. Yang**, 2024: regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6. *Journal of Open Source Software*, **9(100)**, doi: 10.21105/joss.06857
- [5] **Yang, L.**, R. Barkan, K. Srinivasan, J.C. McWilliams, C.J. Shakespeare, and A.H. Gibson, 2023: Oceanic eddies induce a rapid formation of an internal wave continuum. *Communications Earth & Environment*, **4(1)**, doi: 10.1038/s43247-023-01137-1
- [4] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2023: Lee waves break eddy saturation of the Antarctic Circumpolar Current. *Geophysical Research Letters*, **50(11)**, doi: 10.1029/2023GL103866.
- [3] Barkan R., K. Srinivasan, **L. Yang**, J.C. McWilliams, J. Gula, and C. Vic, 2021: Oceanic mesoscale eddy depletion catalyzed by internal waves. *Geophysical Research Letters*, **48(18)**, doi: 10.1029/2021GL094376.
- [2] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2021: The impact of lee waves on the Southern Ocean circulation. *Journal of Physical Oceanography*, **51(9)**, 2933-2950, doi: 10.1175/JPO-D-20-0263.1.
- [1] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2018: Energy Loss from Transient Eddies due to Lee Wave Generation in the Southern Ocean. *Journal of Physical Oceanography*, **48(12)**, 2867–2885, doi: 10.1175/JPO-D-18-0077.1.

Presentations

Invited talks

- [7] **Yang, L.**, et al. Propagating internal-tide-induced wave stresses resolve discrepancies in ocean tide energetics. *Gordon Research Seminar on Ocean Mixing*, South Hadley, Massachusetts, United States, 13-14 June 2026.
- [6] **Yang, L.**, C. J. Shakespeare, B. K. Arbic. Propagating internal-tide-induced wave stresses resolve discrepancies in ocean tide energetics. *School of Marine Sciences, Sun Yat-sen University*, 6 January 2026.
- [5] **Yang, L.**, C. J. Shakespeare, A. K. Morrison, A. M. Hogg, A. H. Gibson, and B. K. Arbic. Finding missing tidal energy dissipation in the open ocean. *Second Institute of Oceanography, MNR*, 17 February 2025.
- [4] **Yang, L.**, R. Barkan, K. Srinivasan, J. C. McWilliams, C. J. Shakespeare, A. H. Gibson. Regulation of internal wave energy and its dissipation by eddies. *MEL Seminar, State Key Laboratory of Marine Environmental Science (Xiamen University)*, 22 February 2023.
- [3] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress. *OCES Departmental Seminar, The Hong Kong University of Science and Technology, Zoom*, 30 September 2022.
- [2] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress. *Physical Oceanography Dissertations Symposium (PODS) XI, Lihue, Kaua'i, Hawaii, United States*, 17-21 October 2021.
- [1] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress. *Polar Oceans seminar, British Antarctic Survey, Zoom*, 4 November 2020.

Selected Oral Presentations

- [14] **Yang, L.**, C. J. Shakespeare, A. K. Morrison, A. M. Hogg, A. H. Gibson, and Brian Arbic. Implementing and evaluating a novel tidal wave drag parameterization in MOM6. *Ocean Modelling and Observations Workshop*, Canberra, ACT, Australia, 2-4 July 2024.
- [13] **Yang, L.**, C. J. Shakespeare, A. K. Morrison, A. M. Hogg, A. H. Gibson, and Brian Arbic. Connecting ocean tides to the large-scale ocean circulation: Tidal parameterization development in MOM6. *Bluelink Annual Science Meeting*, Melbourne, Victoria, Australia, 19-20 March 2024.
- [12] **Yang, L.**, R. Barkan, K. Srinivasan, J. C. McWilliams, C. J. Shakespeare, A. H. Gibson. Oceanic eddies induce a rapid formation of an internal wave continuum. CASPO seminar, Scripps Institution of Oceanography, La Jolla, California, United States, 28 February 2024.
- [11] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. Lee waves break eddy saturation of the Antarctic Circumpolar Current. *Ocean Sciences Meeting 2024*, HE41B-07, New Orleans, Louisiana, United States, 18-23 February 2024.
- [10] **Yang, L.**, R. Barkan, K. Srinivasan, J. C. McWilliams, C. J. Shakespeare, A. H. Gibson. Oceanic eddies induce a rapid formation of an internal wave continuum. *Ocean Sciences Meeting 2024*, PS21A-07, New Orleans, Louisiana, United States, 18-23 February 2024.
- [9] **Yang, L.**, R. Barkan, K. Srinivasan, and J. C. McWilliams. On the propagation of wind-generated near-inertial waves: the importance of submesoscale currents. *Ocean Sciences Meeting 2022, PS11, The dynamics of interacting internal waves and (sub)mesoscale flows*, Virtual, 28 February - 4 March 2022.
- [8] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the response of the Southern Ocean circulation to changes in wind stress. *Second California Geophysical Fluid Dynamics (CalGFD) Meeting*, Zoom, 20-21 August 2020.

- [7] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. Impacts of Lee Waves on the Southern Ocean Circulation and its Sensitivity to Wind Stress. *22nd Conference on Atmospheric and Oceanic Fluid Dynamics (AOFD)*, Idealized Model Approaches to the Atmosphere and Ocean Circulation 7.2, Portland, Maine, United States, 24-28 June 2019.
- [6] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Impacts of Bottom Drag on the Sensitivity of the Southern Ocean Circulation to the Changing Wind. *2018 Consortium for Ocean Sea Ice Modelling in Australia (COSIMA) workshop*, Canberra, Australia, 7-8 May 2018.
- [5] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *2018 Ocean Sciences Meeting*, PO12A-04, Portland, Oregon, United States, 11-16 February 2018.
- [4] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Impacts of Bottom Drag on the Sensitivity of the Southern Ocean Circulation to the Changing Wind. *Joint AMOS National Conference and the International Conference on Southern Hemisphere Meteorology and Oceanography (AMOS-ICSHMO 2018)*, Sydney, Australia, 5-9 February 2018.
- [3] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *Australian Research Council's Centre of Excellence for Climate System Science (ARCCSS) 2017 Workshop*, Canberra, Australia, 30 October - 2 November 2017.
- [2] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *21st Conference on Atmospheric and Oceanic Fluid Dynamics (AOFD) and the 19th Conference on Middle Atmosphere, Mesoscale Ocean Dynamics* 13.2, Portland, Oregon, United States, 26-30 June 2017.
- [1] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *AMOS/MSNZ Conference & ANZ Climate Forum 2017*, Canberra, Australia, 7-10 February 2017.

Selected Honours and Awards

- Best Early Career Researcher (ECR) presentation, Ocean Modelling and Observations Workshop, Canberra, ACT, Australia, 2024.
- Travel grant, Physical Oceanography Dissertations Symposium (PODS) XI, Lihue, Kaua'i, Hawaii, United States, 2021.
- Bursary (€1000), 2017 Summer School on the Fluid Dynamics of Sustainability and the Environment (FDSE), Ecole Polytechnique, Palaiseau, France, 2017.
- Student Oral Presentation Honourable Mention, AMOS/MSNZ Conference & ANZ Climate Forum, Canberra, Australia, 2017.
- Excellent Graduates in the Class of 2015, Ocean University of China, Qingdao, China, 2015.
- National Scholarship (¥8000), Ministry of Education of China, 2014.

Teaching Experience

- Lecturing: Fundamentals of Climate Science (EMSC2021), Australian National University. Semester 2, 2024. Convenor: Dr. Callum Shakespeare.
- Marking: Dynamical Oceanography (KSM310), University of Tasmania. 2017-2019. Coordinators: Dr. Maxim Nikurashin, Prof. Neil Holbrook, Prof. Nathan Bindoff.
- Marking: Advanced Oceanography (KSA306), University of Tasmania. 2016. Coordinator: Dr. Maxim Nikurashin
- Demonstrating and marking: Our Changing Climate (KGA320), University of Tasmania. 2016. Coordinator: Prof. Neil Holbrook

Supervision Experience

- Co-supervision: Anupiya Ellepola, Ph.D., Australian National University (2024-).
- Supervision: Audrey Smart, Ph.B., Australian National University (2024).

Field Experience

- *CTD watch stander*, leg 2 of the GO-SHIP P15S Hydrographic Section, Wellington (NZ) - Lautoka (Fiji), South Pacific, Monitoring Ocean Change and Variability along 170°W from the ice edge to the equator, R/V Investigator, May-June 2016 (35 days).

Professional Services

- *Collaborator*, ACCESS-OM3 development, 2025 - present.
- *Reviewer*, Journal of Physical Oceanography, Journal of Geophysical Research Oceans, Journal of Fluid Mechanics, Journal of Advances in Modeling Earth Systems, Geophysical Research Letters.
- *Group meeting coordinator*, Climate and Fluid Physics, Research School of Earth Sciences, College of Science, ANU, Januray 2024 - December 2025.
- *Discussion leader*, Gordon Research Seminar on Ocean Mixing, South Hadley, Massachusetts, United States, June 2024.
- *Postdoctoral Fellow Selection Committee Member*, Australian National University, March 2024.
- *UTAS PhD student representative* and *Early Career Researcher (ECR) Committee member*, ARC Centre of Excellence for Climate Extremes, 2018.
- *Session Chair*, ARC Centre of Excellence for Climate Extremes Annual Workshop, November 2018.
- *Volunteer*, 22nd Conference on Atmospheric and Oceanic Fluid Dynamics (AOFD), June 2018.

Outreach

- *Volunteer*, ANU open day, 29 March 2025
 - Demonstrated and engaged in a lab experiment with prospective students and their parents on ‘Icebergs in the Southern Ocean’ using a rotating tank (Weather in a Tank Apparatus) and a thermal camera.
- *Volunteer*, IMAS Work Experience program, June-July 2018.
 - Designed and delivered an 1-hour lecture on ‘How do we study oceans? - Research in Physical Oceanography’ for two groups of high school students across Australia;
 - Demonstrated and engaged in a lab experiment with high school students on ‘Ocean gyres (Western Intensification phenomenon)’ using a rotating tank (Weather in a Tank Apparatus).
- *Volunteer*, Young Antarctic Scientists program, May and December 2018.
 - Technical session, 13 December. Engaged in an outdoor practical session, in which year 4 students at Clarendon Vale Primary School observed and learned how to fly a drone.
 - Biogeochemical oceanography session, 13 December. Coordinated a group activity in which year 4 students at Clarendon Vale Primary School used baking soda and vinegar to blow the balloons with CO₂.
 - Physical Oceanography session, 16 May. Engaged in a lecture and an experiment, in which oil, water, honey, vinegar are slowly added into and settled in a tube, aiming to deliver the idea that ‘ocean has layers’, with year 4 students at Clarendon Vale Primary School.